RETAIL BILLING WITH MULTIPLE DISCOUNT RULES

1. Problem Statement

Develop a retail billing system that processes multiple products and quantities. The system calculates product subtotals, applies category-specific taxes, provides discounts when purchase thresholds are reached, and ensures that the total discount does not exceed a maximum concession limit. Finally, it generates an itemized invoice showing the amount payable.

2. Objective

To prepare an accurate retail invoice by calculating product-wise costs, category taxes, threshold-based discounts, maximum discount limits, and the final payable amount.

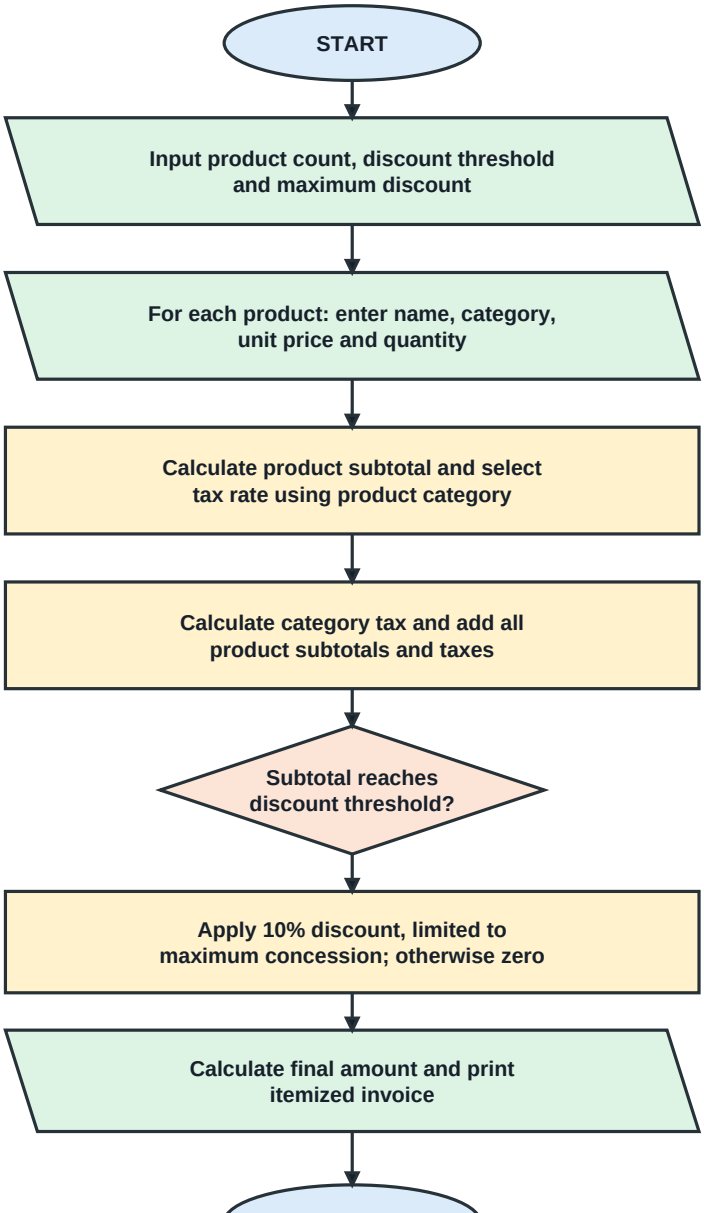
3. Assumptions

1. Product prices, tax rates, discount threshold, and maximum concession are sample values.
2. Each product belongs to one category: food, clothing, or electronics.
3. Tax is calculated on each product subtotal before discount.
4. A 10% discount is applied when the total pre-discount bill reaches the specified threshold.
5. The discount cannot exceed the maximum concession amount.

4. Algorithm

1. Start.
2. Read the number of products, discount threshold, and maximum concession.
3. For each product, read name, category, unit price, and quantity.
4. Calculate the product subtotal.
5. Select the tax rate based on the product category and calculate tax.
6. Add subtotals and taxes to the bill totals.
7. If the subtotal reaches the threshold, calculate the discount.
8. Limit the discount to the maximum concession amount.
9. Calculate the final payable amount.
10. Display the itemized invoice and stop.

5. Flowchart (Clear Normal Symbols)



6. Pseudocode

START
INPUT product_count, threshold, maximum_discount
subtotal_total = 0
tax_total = 0
FOR each product
  INPUT name, category, price, quantity
  subtotal = price * quantity
  SELECT tax_rate using category
  tax = subtotal * tax_rate / 100
  subtotal_total = subtotal_total + subtotal
  tax_total = tax_total + tax
  STORE product details
END FOR
bill_before_discount = subtotal_total + tax_total
IF subtotal_total >= threshold THEN
  discount = subtotal_total * 10 / 100
  IF discount > maximum_discount THEN
    discount = maximum_discount
  END IF
ELSE
  discount = 0
END IF
final_amount = bill_before_discount - discount
DISPLAY itemized invoice and final_amount
STOP

7. Python Code

```
print("RETAIL BILLING SYSTEM")

products = []
product_count = int(input("Enter number of products: "))
threshold = float(input("Enter discount threshold: "))
maximum_discount = float(input("Enter maximum discount amount: "))

subtotal_total = 0
tax_total = 0

for i in range(product_count):
    print("\\nProduct", i + 1)
    name = input("Enter product name: ")
    category = input("Enter category (food/clothing/electronics): ").lower()
    price = float(input("Enter unit price: "))
    quantity = int(input("Enter quantity: "))

    subtotal = price * quantity

    if category == "food":
        tax_rate = 5
    elif category == "clothing":
        tax_rate = 12
    else:
        tax_rate = 18

    tax = subtotal * tax_rate / 100
    subtotal_total = subtotal_total + subtotal
    tax_total = tax_total + tax

    products.append([name, category, price, quantity,
                     subtotal, tax_rate, tax])

bill_before_discount = subtotal_total + tax_total

if subtotal_total >= threshold:
    discount = subtotal_total * 10 / 100
    if discount > maximum_discount:
        discount = maximum_discount
else:
    discount = 0

final_amount = bill_before_discount - discount

print("\\n---------- ITEMIZED INVOICE ----------")
for item in products:
    print("Product:", item[0])
    print("Category:", item[1])
    print("Quantity:", item[3])
    print("Subtotal: Rs.", round(item[4], 2))
    print("Tax:", item[5], "%")
    print("Tax amount: Rs.", round(item[6], 2))
    print("--------------------------------------")

print("Subtotal: Rs.", round(subtotal_total, 2))
print("Total tax: Rs.", round(tax_total, 2))
print("Discount: Rs.", round(discount, 2))
print("Final amount: Rs.", round(final_amount, 2))
```

8. Expected Output

```
RETAIL BILLING SYSTEM
Enter number of products: 2
Enter discount threshold: 1000
Enter maximum discount amount: 150

Product 1
Enter product name: Shirt
Enter category: clothing
Enter unit price: 500
Enter quantity: 2

Product 2
Enter product name: Rice
Enter category: food
Enter unit price: 300
Enter quantity: 1

---------- ITEMIZED INVOICE ----------
Product: Shirt | Quantity: 2
Subtotal: Rs. 1000.0 | Tax: 12% | Tax amount: Rs. 120.0
Product: Rice | Quantity: 1
Subtotal: Rs. 300.0 | Tax: 5% | Tax amount: Rs. 15.0
Subtotal: Rs. 1300.0
Total tax: Rs. 135.0
Discount: Rs. 130.0
Final amount: Rs. 1305.0
```